TASK: 1

Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

I opened Wireshark and selected my active Wi-Fi network interface. I started capturing network traffic, opened the Flipkart homepage, and refreshed the page. After a few seconds, I stopped the capture and examined the Protocol and Info columns to identify packets belonging to the four TCP/IP layers.

Sr. No.	TCP/IP Layer	Example Protocol	Packet Information	Function
1	Application	DNS	DNS query for a website domain	Converts a domain name into an IP address.
2	Transport	TCP	TCP packet containing source and destination ports	Provides reliable data transmission.
3	Internet	IPv4 or IPv6	IP packet containing source and destination IP addresses	Routes packets between networks.
4	Network Access	Ethernet or Wi-Fi (802.11)	Frame containing link-layer addresses	Transfers data over the local network.

Conclusion: Wireshark helps us examine network packets and understand how the TCP/IP layers work when we access a website such as Flipkart.

TASK: 2

Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

The OSI model has seven layers, whereas the TCP/IP model is commonly represented using four layers.

OSI Model Layer	TCP/IP Model Layer	Function	Real-World Example
7. Application	Application	Provides network services to applications.	Sending a message on WhatsApp
6. Presentation	Application	Handles data formatting, encryption, and encoding.	Encrypting WhatsApp messages
5. Session	Application	Establishes and manages communication sessions.	Maintaining a video-call session
4. Transport	Transport	Provides end-to-end data delivery.	TCP or UDP carrying application data
3. Network	Internet	Handles IP addressing and packet routing.	Routing a Zomato order request to a server
2. Data Link	Network Access	Transfers frames over a local network.	Sending data through Wi-Fi
1. Physical	Network Access	Transmits bits using physical or wireless signals.	Wi-Fi radio signals or Ethernet cables

Main Differences:

1. The OSI model has seven layers, while the TCP/IP model commonly has four.
2. The OSI model is mainly a conceptual reference model for understanding network communication.
3. The TCP/IP model is based on the protocols used for communication over the internet.
4. The OSI Application, Presentation, and Session layers are combined into the TCP/IP Application layer.
5. The OSI Data Link and Physical layers are combined into the TCP/IP Network Access layer.

Conclusion: Both models explain how data travels between devices, but the TCP/IP model more directly represents internet communication.

TASK: 3

Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.

A TCP segment header contains important fields that help devices establish connections, identify applications, maintain the correct data order, and control data transmission.

Sr. No.	TCP Header Field	Function
1	Source Port	Identifies the sending application's communication endpoint.
2	Destination Port	Identifies the destination application's communication endpoint.
3	Sequence Number	Identifies the position of data in the TCP byte stream.
4	Acknowledgment Number	Indicates the next byte expected from the other device when the ACK flag is set.
5	Data Offset (Header Length)	Specifies the length of the TCP header.
6	Reserved Bits	Reserved for protocol use and normally set as specified by the TCP standard.
7	Flags (Control Bits)	Indicate connection and data-control functions, such as SYN, ACK, FIN, and RST.
8	Window Size	Indicates how much data the receiver is currently willing to accept.
9	Checksum	Helps detect corruption in the TCP header and data.
10	Urgent Pointer	Identifies urgent data when the URG flag is set.
11	Options	Carries additional TCP parameters, such as Maximum Segment Size (MSS).

12	Padding	Adds bits to align the TCP header to a 32-bit boundary.

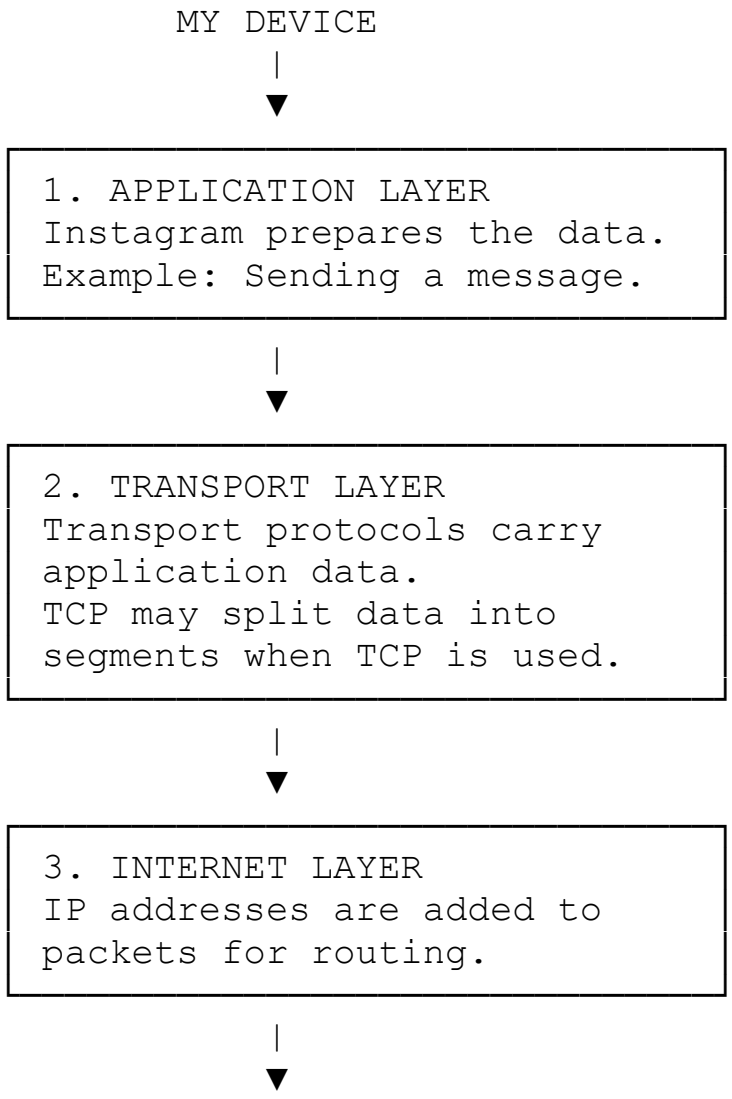
Conclusion: TCP header fields help ensure that data is delivered to the correct application, received in the correct order, and transmitted reliably.

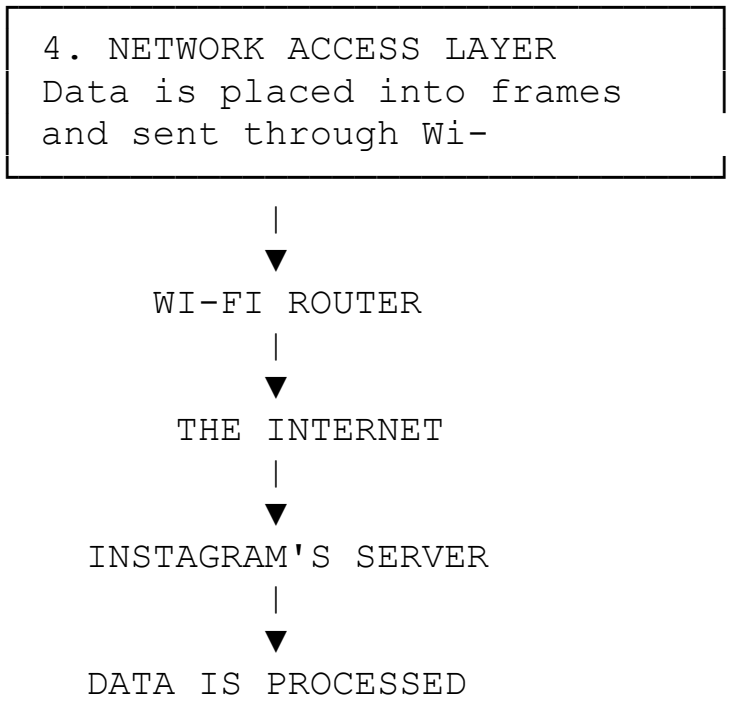
TASK: 4.

Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

When I send a message or upload a photo to Instagram, the data passes through the four TCP/IP layers before reaching Instagram's server.

TCP/IP Model Data Flow





Explanation:

1. **Application Layer:** I type a message or select a photo in Instagram.
2. **Transport Layer:** The appropriate transport protocol carries the application data. TCP divides the byte stream into segments when TCP is used.
3. **Internet Layer:** IP adds source and destination IP addresses so that packets can travel across networks.
4. **Network Access Layer:** The data is placed into frames and transmitted through Wi-Fi or another network connection.

When the data reaches Instagram's server, the server processes it and performs the requested action.

Conclusion: The TCP/IP model explains how application data is prepared, transported, routed, and transmitted across a network.

TASK: 5

Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

I selected a TCP packet from my Wireshark capture while refreshing the Flipkart homepage.

Item	Observation
Selected Protocol	TCP
TCP/IP Layer	Transport Layer
How I Identified It	I checked the Protocol column and expanded the TCP section in the packet details pane.
Header Information	Source Port, Destination Port, Sequence Number, and Acknowledgment Number, depending on the selected packet
Real-World Action	The packet represents part of the network communication that occurs when the browser loads the Flipkart homepage.

Explanation:

TCP belongs to the Transport layer of the TCP/IP model. I identified the packet by examining the Protocol column in Wireshark and checking the TCP header fields. The packet represents part of the communication between my browser and a remote service while the Flipkart webpage is loading.

Conclusion: Wireshark allows us to identify network protocols and understand how applications communicate with remote servers.